

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-OXY-004

Title: Safe Handling of Oxygen Pipelines and Cylinders

Department: Oxygen Plant

Revision: 02

Date: 19-Sep-2024

1. PURPOSE

To prevent fires caused by oxygen enrichment and to ensure safe handling of oxygen pipelines and cylinders.

2. KEY HAZARD

Oxygen does not burn, but it makes everything else burn much faster. Clothing soaked in oxygen can catch fire from a small spark. Oxygen-enriched air is above 23.5% oxygen.

3. RULES FOR OXYGEN PIPELINES

- Oil and grease must NEVER come into contact with oxygen valves, fittings or regulators
- Use only oxygen-cleaned and degreased tools
- Open oxygen valves slowly; never open a valve fast
- Maximum velocity in carbon steel oxygen lines: 8 m/s
- Before hot work near an oxygen line, confirm oxygen in air is below 23.5%

4. OXYGEN CYLINDERS

- Store cylinders upright, chained, and at least 6 metres away from fuel gas cylinders (or separated by a fire wall)
- Cylinder colour code: black body with white shoulder
- Use a trolley to move cylinders; never roll or drag them

5. IF CLOTHING IS OXYGEN-ENRICHED

- Leave the area, do not smoke or go near any ignition source
- Air out clothing for at least 15 minutes before returning

6. EMERGENCY

On an oxygen leak or fire, isolate the supply valve if safe, evacuate, and call the Fire Station: Ext. 101 and Emergency Control Room: Ext. 100.